ECON 0100 | Vignette C2 | Taxes + Subsidies

Solutions

Answers and work are shown in the red blocks.

Before the connection between Glittering Gum and Starlight Sneeze was established, the fiscal arm of the Ministry implemented a 3 Coin subsidy on every sale of Glittering Gum, to put a little sparkle into Hogsmeade's conversations. The Demand and Supply curves are the same as in Vignette C1.

With no known externalities, use a graph to illustrate the impact of the subsidy on the market.

Buyer's Price (P_b): _____ Seller's Price (P_s): _____

Subsidy Quantity (Q^{Sub}): _____ Dead Weight Loss (DWL): _____

Buyer's Price P_b: 12 coins. Seller's Price P_s: 15 coins. Subsidy Quantity Q^{Sub}: 36 pieces of gum. Dead Weight Loss: 6 coins.

From Vignette C1, demand is $P = 30 - \frac{1}{2}Q$ and supply is $P = 6 + \frac{1}{4}Q$. Without the subsidy, buyers and sellers face one price:

$$30 - \tfrac{1}{2}Q = 6 + \tfrac{1}{4}Q \quad \Rightarrow \quad Q = 32, \quad P = 14$$

A subsidy is the tax wedge turned around: the seller now receives 3 more than the buyer pays, so the wedge sits to the *right* of the no-policy equilibrium:

$$P_s = 3 + P_b \quad \Rightarrow \quad 6 + \tfrac{1}{4}Q = 3 + (30 - \tfrac{1}{2}Q) \quad \Rightarrow \quad Q =$$

so $P_b = 30 - 18 = 12$ and $P_s = 6 + 9 = 15$.

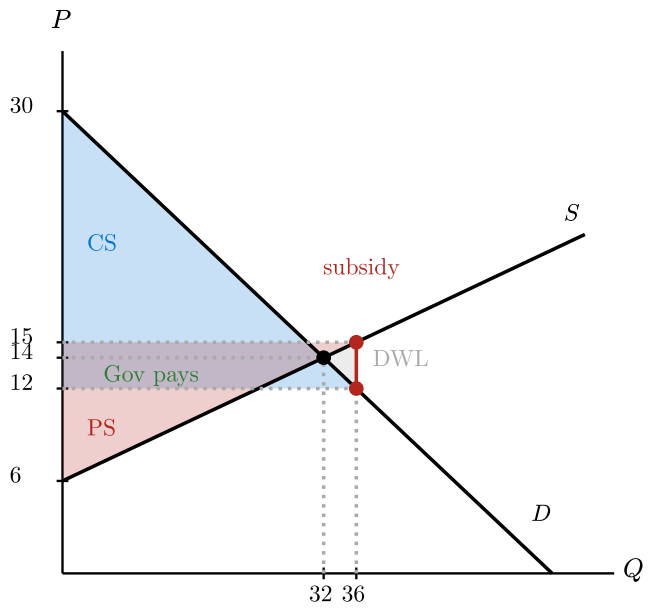


Figure 1: Subsidy equilibrium

Past 32, supply (marginal cost) is above demand (marginal benefit): the 33rd through 36th pieces cost 14.25 to 15 coins to make, but buyers value them at 14 down to 12. They trade only because

the Ministry pays the difference. The triangle between the curves over those 4 pieces is the deadweight loss:

$$\text{DWL} = \tfrac{1}{2} \times 3 \times (36 - 32) = 6$$

For reference, since the graph shows them: buyers pay $14 - 12 = 2$ less than before and sellers receive $15 - 14 = 1$ more, so buyers get more of the subsidy (demand is steeper). Consumer surplus rises from $\frac{1}{2} \times 32 \times 16 = 256$ to $\frac{1}{2} \times 36 \times 18 = 324$ and producer surplus from $\frac{1}{2} \times 32 \times 8 = 128$ to $\frac{1}{2} \times 36 \times 9 = 162$, but the Ministry pays $3 \times 36 = 108$. Check: the gains, $68 + 34 = 102$, are the spending less the deadweight loss, $108 - 6$.

With no externality, the market's 32 was already efficient, so the subsidy buys 4 pieces too many. Once the Starlight Sneeze is counted (Vignette C1), the efficient quantity is 24, so the Ministry was paying to push the market the wrong way; Vignette C3 takes that up.